

Calculating statistics

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Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
library(limma)
library(edge)
library(genefilter)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase", "limma", "genefilter", "jdstorey/edge"))
```

Download the data

Here we are going to use some data from the paper Evaluating gene expression in C57BL/6J and DBA/2J mouse striatum using RNA-Seq and microarrays. (<http://www.ncbi.nlm.nih.gov/pubmed?term=21455293>) that is a comparative RNA-seq analysis of different mouse strains.

```
con = url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bottomly_eset.RData")
load(file=con)
close(con)
bot = bottomly.eset
pdata=pData(bot)
edata=as.matrix(exprs(bot))
fdata = fData(bot)
ls()
```

```
## [1] "bot"          "bottomly.eset" "con"          "edata"  
## [5] "fdata"        "pdata"         "tropical"
```

Transform the data

Here we will transform the data and remove lowly expressed genes.

```
edata = log2(as.matrix(edata) + 1)  
edata = edata[rowMeans(edata) > 10, ]
```

Calculate t- or F-statistics rapidly

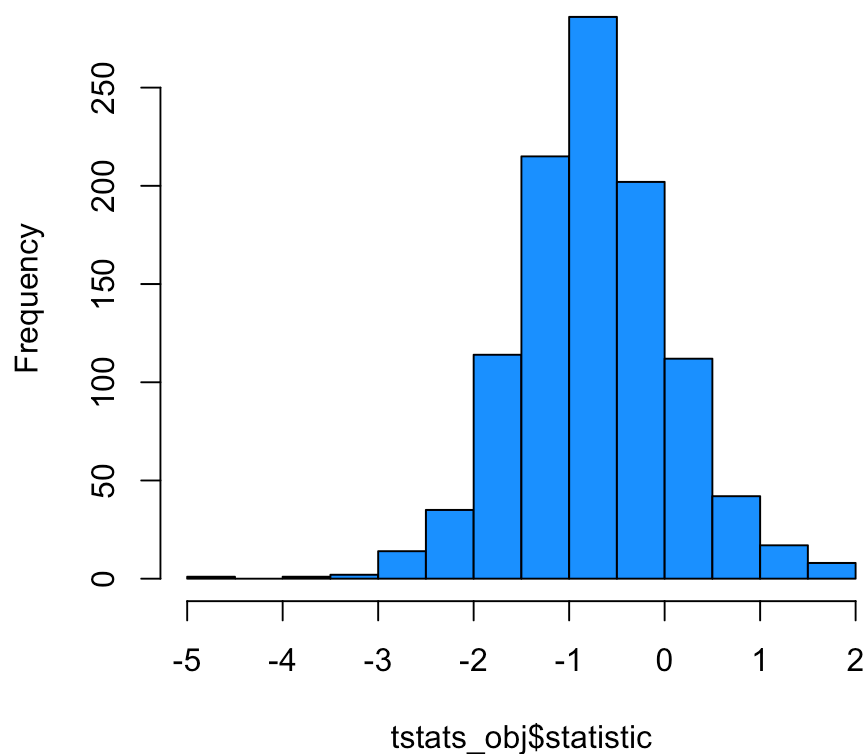
The `genefilter` package lets you compute statistics rapidly for very simple cases (two or multi-group) comparisons. These are not moderated in any way.

```
tstats_obj = rowttests(edata,pdata$strain)  
names(tstats_obj)
```

```
## [1] "statistic" "dm"          "p.value"
```

```
hist(tstats_obj$statistic,col=2)
```

Histogram of `tstats_obj$statistic`



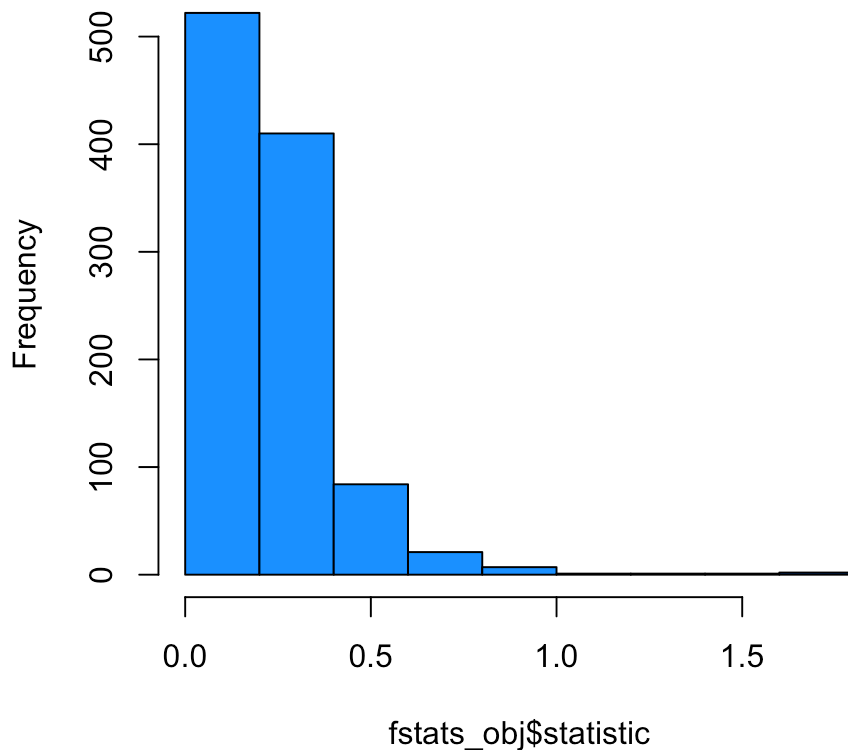
For multi-group comparisons use the F-statistic

```
fstats_obj = rowFtests(edata,as.factor(pdata$lane.number))  
names(fstats_obj)
```

```
## [1] "statistic" "p.value"
```

```
hist(fstats_obj$statistic,col=2)
```

Histogram of fstats_obj\$statistic



Fit many statistics with limma

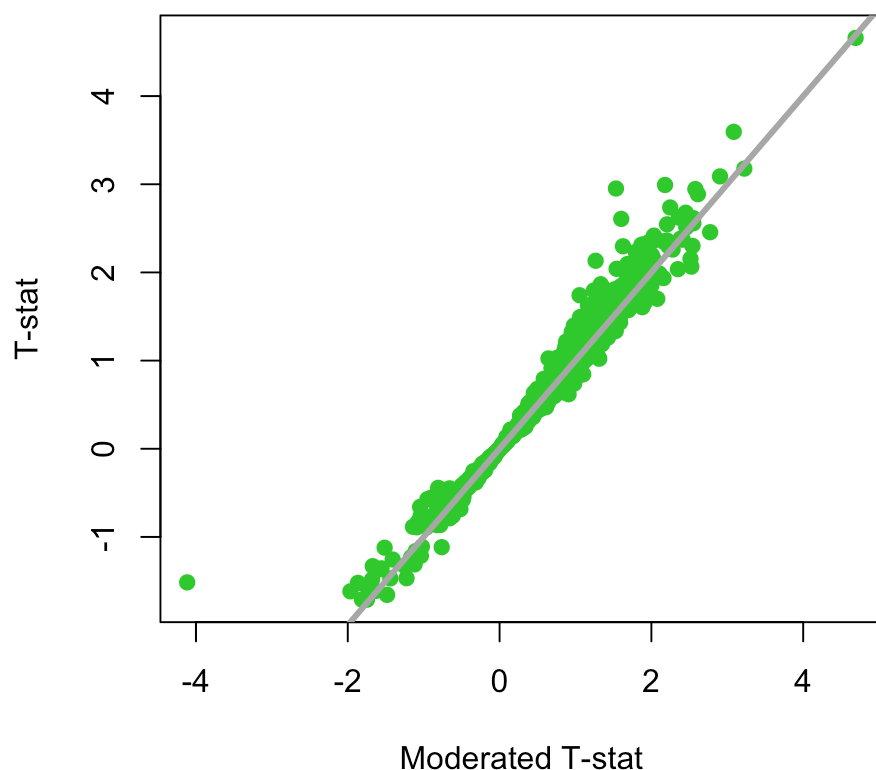
This approach fits many moderated statistics simultaneously

```
mod = model.matrix(~ pdata$strain)  
fit_limma = lmFit(edata,mod)  
ebayes_limma = eBayes(fit_limma)  
head(ebayes_limma$t)
```

```
##               (Intercept) pdata$strainDBA/2J
## ENSMUSG000000000131    61.56232      1.48876396
## ENSMUSG000000000149    63.84978     -0.09514821
## ENSMUSG000000000374    63.48896      1.00196514
## ENSMUSG000000000399    60.07990      0.19593890
## ENSMUSG000000000532    64.97678      0.80640805
## ENSMUSG000000000605    68.18274      0.79136446
```

If the model is unadjusted you get the moderated version of the statistic from `rowttests`

```
plot(ebayes_limma$t[,2], -tstats_obj$statistic, col=4,
     xlab="Moderated T-stat", ylab="T-stat")
abline(c(0,1), col="darkgrey", lwd=3)
```



Fit many adjusted statistics with limma

Here we adjust for the lane number, now the test-statistic for the strain is adjusted for the lane number (a surrogate for a batch effect).

```
mod_adj = model.matrix(~ pdata$strain + as.factor(pdata$lane.number))
fit_limma_adj = lmFit(edata, mod_adj)
ebayes_limma_adj = eBayes(fit_limma_adj)
head(ebayes_limma_adj$t)
```

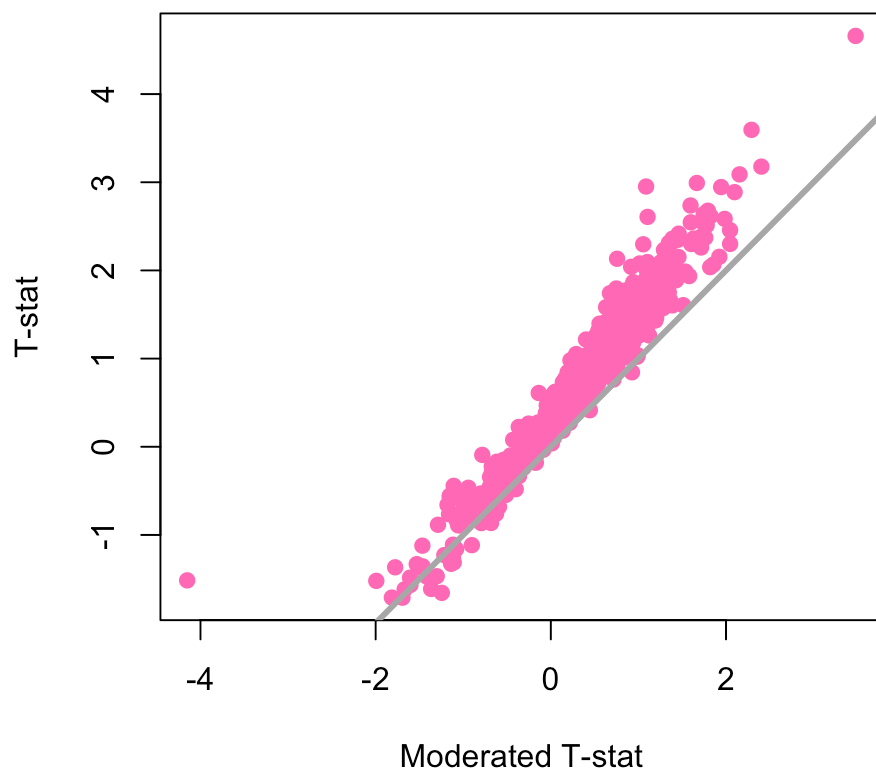
```

## (Intercept) pdata$strainDBA/2J
## ENSMUSG00000000131 27.42608 0.993341906
## ENSMUSG00000000149 28.30319 -0.302293604
## ENSMUSG00000000374 28.43334 0.684161612
## ENSMUSG00000000399 26.89729 -0.007509614
## ENSMUSG00000000532 29.20894 0.552296098
## ENSMUSG00000000605 30.52908 0.450401742
## as.factor(pdata$lane.number)2
## ENSMUSG00000000131 0.106103723
## ENSMUSG00000000149 0.283169296
## ENSMUSG00000000374 0.062660205
## ENSMUSG00000000399 0.066998156
## ENSMUSG00000000532 -0.036205851
## ENSMUSG00000000605 0.001102388
## as.factor(pdata$lane.number)3
## ENSMUSG00000000131 0.04870234
## ENSMUSG00000000149 0.15268674
## ENSMUSG00000000374 -0.04788353
## ENSMUSG00000000399 -0.13957244
## ENSMUSG00000000532 -0.13080316
## ENSMUSG00000000605 -0.08760933
## as.factor(pdata$lane.number)5
## ENSMUSG00000000131 0.137709189
## ENSMUSG00000000149 0.114049004
## ENSMUSG00000000374 -0.031272733
## ENSMUSG00000000399 0.156240669
## ENSMUSG00000000532 -0.007169364
## ENSMUSG00000000605 0.185393375
## as.factor(pdata$lane.number)6
## ENSMUSG00000000131 0.7274749
## ENSMUSG00000000149 1.0154366
## ENSMUSG00000000374 0.5228153
## ENSMUSG00000000399 0.8271336
## ENSMUSG00000000532 0.3135213
## ENSMUSG00000000605 0.4940291
## as.factor(pdata$lane.number)7
## ENSMUSG00000000131 0.41722714
## ENSMUSG00000000149 0.58175790
## ENSMUSG00000000374 0.23222225
## ENSMUSG00000000399 0.05475727
## ENSMUSG00000000532 0.14890971
## ENSMUSG00000000605 0.27370736
## as.factor(pdata$lane.number)8
## ENSMUSG00000000131 0.28886043
## ENSMUSG00000000149 0.39833581
## ENSMUSG00000000374 0.11020429
## ENSMUSG00000000399 -0.01182217
## ENSMUSG00000000532 -0.02978295
## ENSMUSG00000000605 0.18942807

```

If the model is adjusted you get the moderated version of the statistic but it doesn't match `rowttests` because it is adjusted

```
plot(ebayes_limma_adj$t[,2],-tstats_obj$statistic,col=3,
     xlab="Moderated T-stat",ylab="T-stat")
abline(c(0,1),lwd=3,col="darkgrey")
```



Calculating a nested model comparison with limma

Sometimes we want to compare the null model to the alternative model with some additional covariates. Here we have to know which coefficients we want to test in the alternative model.

Suppose we wanted to find lane effects then we can fit a limma model and find which coefficients belong to the lane variable.

```
mod_lane = model.matrix(~ as.factor(pdata$lane.number))
fit_limma_lane = lmFit(edata,mod_lane)
ebayes_limma_lane = eBayes(fit_limma_lane)
head(ebayes_limma_lane$t)
```

```
## (Intercept) as.factor(pdata$lane.number)2
## ENSMUSG00000000131 29.18101 0.108054558
## ENSMUSG00000000149 29.75296 0.288375678
## ENSMUSG00000000374 30.15864 0.063812283
## ENSMUSG00000000399 28.35119 0.068229991
## ENSMUSG00000000532 30.94034 -0.036871536
## ENSMUSG00000000605 32.30420 0.001122656
## as.factor(pdata$lane.number)3
## ENSMUSG00000000131 0.04959779
## ENSMUSG00000000149 0.15549405
## ENSMUSG00000000374 -0.04876393
## ENSMUSG00000000399 -0.14213864
## ENSMUSG00000000532 -0.13320812
## ENSMUSG00000000605 -0.08922013
## as.factor(pdata$lane.number)5
## ENSMUSG00000000131 0.33389895
## ENSMUSG00000000149 0.06002328
## ENSMUSG00000000374 0.09926018
## ENSMUSG00000000399 0.16048443
## ENSMUSG00000000532 0.09886277
## ENSMUSG00000000605 0.27882684
## as.factor(pdata$lane.number)6
## ENSMUSG00000000131 0.9451392
## ENSMUSG00000000149 0.9942322
## ENSMUSG00000000374 0.6735237
## ENSMUSG00000000399 0.8558060
## ENSMUSG00000000532 0.4312304
## ENSMUSG00000000605 0.5987006
## as.factor(pdata$lane.number)7
## ENSMUSG00000000131 0.62359473
## ENSMUSG00000000149 0.54476241
## ENSMUSG00000000374 0.37234955
## ENSMUSG00000000399 0.05530581
## ENSMUSG00000000532 0.26062501
## ENSMUSG00000000605 0.37035652
## as.factor(pdata$lane.number)8
## ENSMUSG00000000131 0.49055391
## ENSMUSG00000000149 0.35466155
## ENSMUSG00000000374 0.24588867
## ENSMUSG00000000399 -0.01369792
## ENSMUSG00000000532 0.07542578
## ENSMUSG00000000605 0.28300845
```

Then we can get the F-statistics with `topTable`

```
top_lane = topTable(ebayes_limma_lane, coef=2:7,
                    number=dim(edata)[1], sort.by="none")
head(top_lane)
```

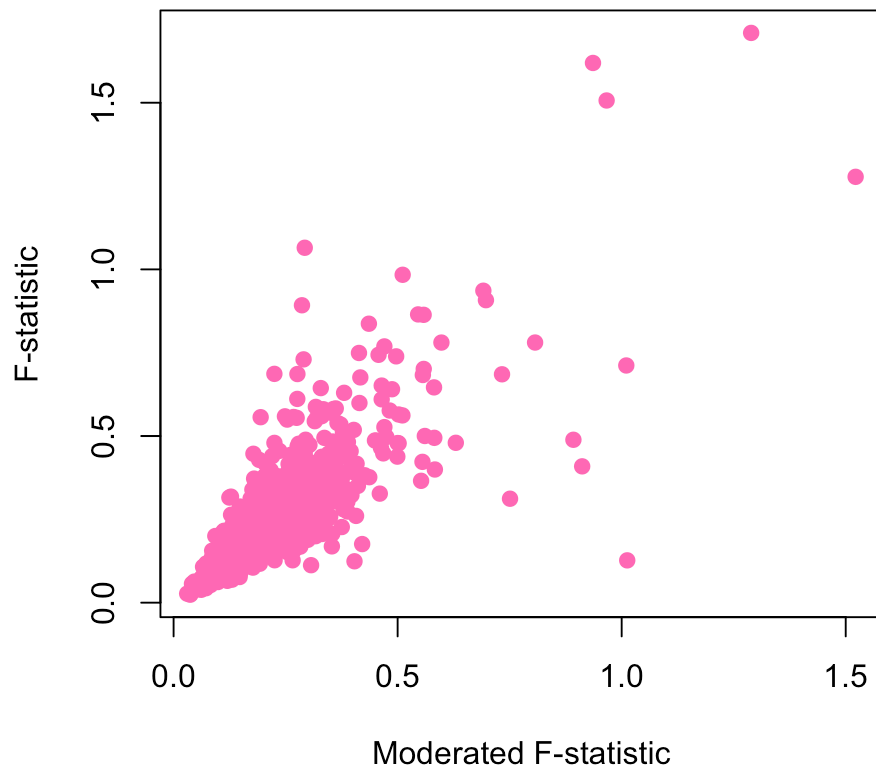
```

##          as.factor.pdata.lane.number.2
## ENSMUSG00000000131          0.0545143060
## ENSMUSG00000000149          0.1454876148
## ENSMUSG00000000374          0.0321937582
## ENSMUSG00000000399          0.0344225239
## ENSMUSG00000000532         -0.0186019566
## ENSMUSG00000000605          0.0005663883
##          as.factor.pdata.lane.number.3
## ENSMUSG00000000131          0.02502244
## ENSMUSG00000000149          0.07844787
## ENSMUSG00000000374         -0.02460175
## ENSMUSG00000000399         -0.07170997
## ENSMUSG00000000532         -0.06720446
## ENSMUSG00000000605         -0.04501220
##          as.factor.pdata.lane.number.5
## ENSMUSG00000000131          0.16845444
## ENSMUSG00000000149          0.03028218
## ENSMUSG00000000374          0.05007748
## ENSMUSG00000000399          0.08096556
## ENSMUSG00000000532          0.04987698
## ENSMUSG00000000605          0.14067016
##          as.factor.pdata.lane.number.6
## ENSMUSG00000000131          0.4768296
## ENSMUSG00000000149          0.5015973
## ENSMUSG00000000374          0.3397976
## ENSMUSG00000000399          0.4317603
## ENSMUSG00000000532          0.2175589
## ENSMUSG00000000605          0.3020488
##          as.factor.pdata.lane.number.7
## ENSMUSG00000000131          0.31460805
## ENSMUSG00000000149          0.27483657
## ENSMUSG00000000374          0.18785304
## ENSMUSG00000000399          0.02790218
## ENSMUSG00000000532          0.13148720
## ENSMUSG00000000605          0.18684754
##          as.factor.pdata.lane.number.8 AveExpr          F
## ENSMUSG00000000131          0.24748800 10.59389 0.23962863
## ENSMUSG00000000149          0.17892932 10.78688 0.23360430
## ENSMUSG00000000374          0.12405261 10.86015 0.12886955
## ENSMUSG00000000399         -0.00691070 10.18493 0.21598161
## ENSMUSG00000000532          0.03805285 11.08784 0.07307893
## ENSMUSG00000000605          0.14277981 11.62820 0.12134214
##          P.Value adj.P.Val
## ENSMUSG00000000131 0.9635205 0.9998831
## ENSMUSG00000000149 0.9657594 0.9998831
## ENSMUSG00000000374 0.9927726 0.9998831
## ENSMUSG00000000399 0.9718839 0.9998831
## ENSMUSG00000000532 0.9985087 0.9998831
## ENSMUSG00000000605 0.9938655 0.9998831

```

This is again the moderated version of the F-statistic we saw earlier

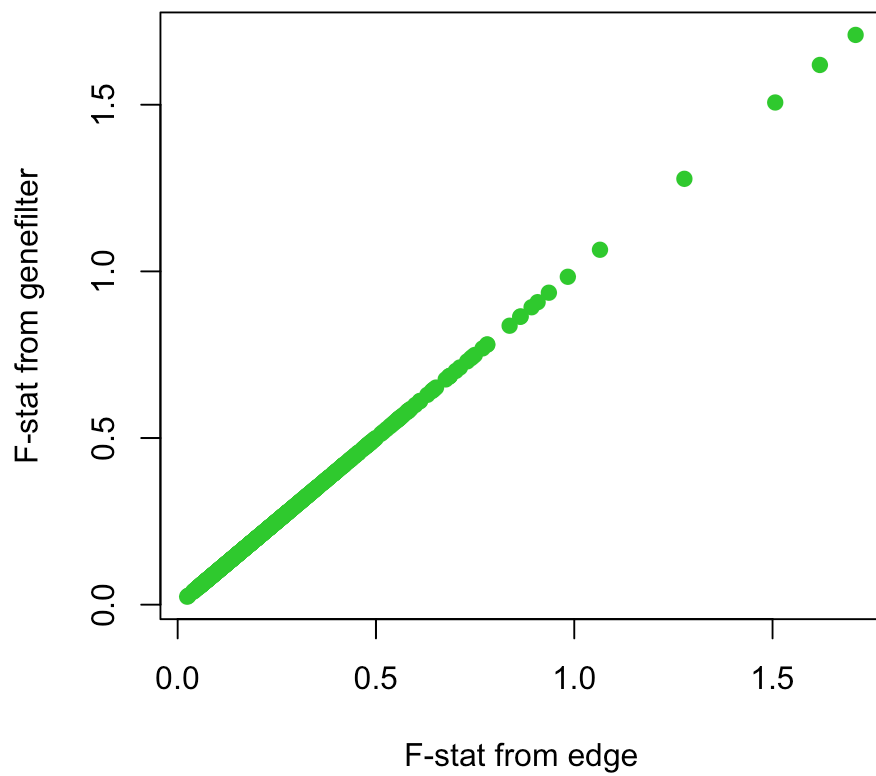

```
plot(top_lane$F,fstats_obj$statistic,
     xlab="Moderated F-statistic",ylab="F-statistic",col=3)
```



Calculating a nested comparison with edge

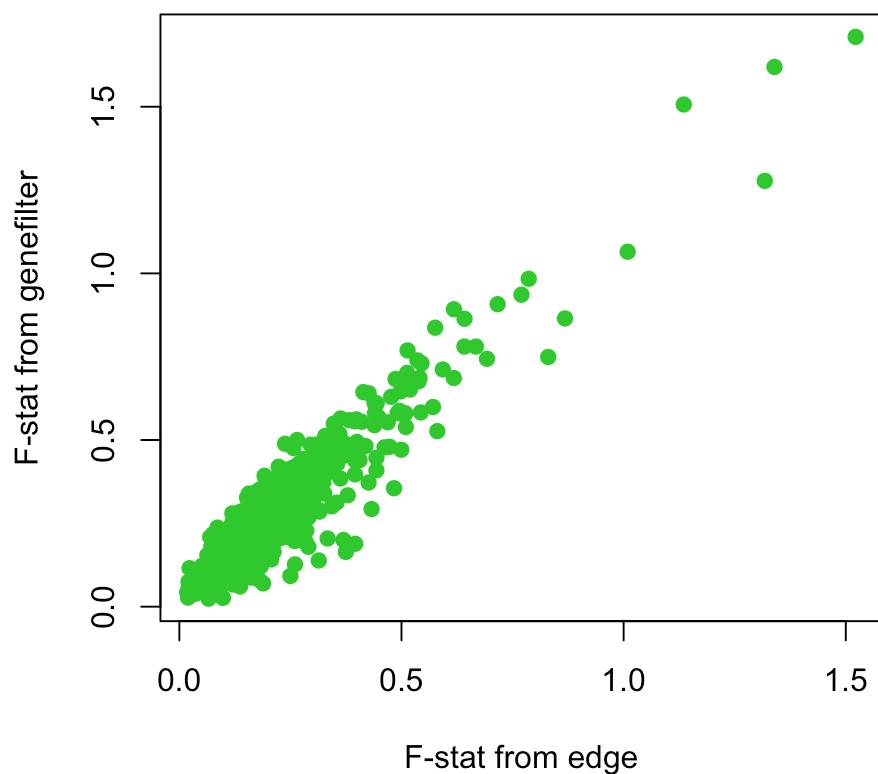
We can also perform the unmoderated nested comparisons in the `edge` package, which also has functions for calculating a more powerful `odp` statistic (http://genomine.org/papers/SDL_Biostat_2007.pdf)

```
edge_study = build_study(edata, grp = as.factor(pdata$lane.number))
de_obj = lrt(edge_study)
qval = qvalueObj(de_obj)
plot(qval$stat,fstats_obj$statistic,col=4,
     xlab="F-stat from edge",ylab="F-stat from genefilter")
```



We can easily adjust for variables by passing arguments to the `adj.var` variable.

```
edge_study2 = build_study(edata, grp = as.factor(pdata$lane.number),
                          adj.var=pdata$strain)
de_obj2 = lrt(edge_study2)
qval2 = qvalueObj(de_obj2)
plot(qval2$stat, fstats_obj$statistic, col=4,
     xlab="F-stat from edge", ylab="F-stat from genefilter")
```



More information

You can find a lot more information on this model fitting strategy in:

- The limma paper
(<http://www.bioconductor.org/packages/release/bioc/vignettes/limma/inst/doc/usersguide.pdf>)
- The limma vignette
(<http://www.bioconductor.org/packages/release/bioc/vignettes/limma/inst/doc/usersguide.pdf>)

Session information

Here is the session information

```
devtools::session_info()
```

```

## setting value
## version R version 3.2.1 (2015-06-18)
## system x86_64, darwin10.8.0
## ui RStudio (0.99.447)
## language (EN)
## collate en_US.UTF-8
## tz America/New_York
##
## package * version date
## acepack 1.3-3.3 2014-11-24
## annotate 1.46.1 2015-07-11
## AnnotationDbi * 1.30.1 2015-04-26
## assertthat 0.1 2013-12-06
## BiasedUrn * 1.06.1 2013-12-29
## Biobase * 2.28.0 2015-04-17
## BiocGenerics * 0.14.0 2015-04-17
## BiocInstaller * 1.18.4 2015-07-22
## BiocParallel 1.2.20 2015-08-07
## biomaRt 2.24.0 2015-04-17
## Biostrings 2.36.3 2015-08-12
## bitops 1.0-6 2013-08-17
## bladderbatch * 1.6.0 2015-08-26
## broom * 0.3.7 2015-05-06
## caTools 1.17.1 2014-09-10
## cluster 2.0.3 2015-07-21
## colorspace 1.2-6 2015-03-11
## corpcor 1.6.8 2015-07-08
## curl 0.9.2 2015-08-08
## DBI * 0.3.1 2014-09-24
## dendextend * 1.1.0 2015-07-31
## DESeq2 * 1.8.1 2015-05-02
## devtools * 1.8.0 2015-05-09
## digest 0.6.8 2014-12-31
## dplyr * 0.4.3 2015-09-01
## edge * 2.1.0 2015-09-06
## evaluate 0.7.2 2015-08-13
## foreign 0.8-66 2015-08-19
## formatR 1.2 2015-04-21
## Formula * 1.2-1 2015-04-07
## futile.logger 1.4.1 2015-04-20
## futile.options 1.0.0 2010-04-06
## gdata 2.17.0 2015-07-04
## genefilter * 1.50.0 2015-04-17
## geneLenDataBase * 1.4.0 2015-09-06
## geneplotter 1.46.0 2015-04-17
## GenomeInfoDb * 1.4.2 2015-08-15
## GenomicAlignments 1.4.1 2015-04-24
## GenomicFeatures 1.20.2 2015-08-14
## GenomicRanges * 1.20.5 2015-06-09
## genstats * 0.1.02 2015-09-05
## ggplot2 * 1.0.1 2015-03-17
## git2r 0.11.0 2015-08-12

```

##	GO.db	3.1.2	2015-09-06
##	goseq	* 1.20.0	2015-04-17
##	gplots	* 2.17.0	2015-05-02
##	gridExtra	2.0.0	2015-07-14
##	gtable	0.1.2	2012-12-05
##	gtools	3.5.0	2015-05-29
##	highr	0.5	2015-04-21
##	HistData	* 0.7-5	2014-04-26
##	Hmisc	* 3.16-0	2015-04-30
##	htmltools	0.2.6	2014-09-08
##	httr	1.0.0	2015-06-25
##	IRanges	* 2.2.7	2015-08-09
##	KernSmooth	2.23-15	2015-06-29
##	knitr	* 1.11	2015-08-14
##	lambda.r	1.1.7	2015-03-20
##	lattice	* 0.20-33	2015-07-14
##	latticeExtra	0.6-26	2013-08-15
##	lazyeval	0.1.10	2015-01-02
##	limma	* 3.24.15	2015-08-06
##	lme4	1.1-9	2015-08-20
##	locfit	1.5-9.1	2013-04-20
##	magrittr	1.5	2014-11-22
##	MASS	* 7.3-43	2015-07-16
##	Matrix	* 1.2-2	2015-07-08
##	MatrixEQTL	* 2.1.1	2015-02-03
##	memoise	0.2.1	2014-04-22
##	mgcv	* 1.8-7	2015-07-23
##	minqa	1.2.4	2014-10-09
##	mnormt	1.5-3	2015-05-25
##	munsell	0.4.2	2013-07-11
##	nlme	* 3.1-122	2015-08-19
##	nloptr	1.0.4	2014-08-04
##	nnet	7.3-10	2015-06-29
##	org.Hs.eg.db	* 3.1.2	2015-07-17
##	plyr	1.8.3	2015-06-12
##	preprocessCore	* 1.30.0	2015-04-17
##	proto	0.3-10	2012-12-22
##	psych	1.5.6	2015-07-08
##	qvalue	* 2.0.0	2015-04-17
##	R6	2.1.1	2015-08-19
##	RColorBrewer	1.1-2	2014-12-07
##	Rcpp	* 0.12.0	2015-07-25
##	RcppArmadillo	* 0.5.400.2.0	2015-08-17
##	RCurl	1.95-4.7	2015-06-30
##	reshape2	1.4.1	2014-12-06
##	rmarkdown	0.7	2015-06-13
##	rpart	4.1-10	2015-06-29
##	Rsamtools	1.20.4	2015-06-01
##	RSkittleBrewer	* 1.1	2015-09-05
##	RSQLite	* 1.0.0	2014-10-25
##	rstudioapi	0.3.1	2015-04-07
##	rtracklayer	1.28.9	2015-08-19

##	rversions	1.0.2	2015-07-13
##	S4Vectors	* 0.6.5	2015-09-01
##	scales	0.3.0	2015-08-25
##	snm	1.16.0	2015-04-17
##	snpStats	* 1.18.0	2015-04-17
##	stringi	0.5-5	2015-06-29
##	stringr	1.0.0	2015-04-30
##	survival	* 2.38-3	2015-07-02
##	sva	* 3.14.0	2015-04-17
##	tidyr	0.2.0	2014-12-05
##	UsingR	* 2.0-5	2015-08-06
##	whisker	0.3-2	2013-04-28
##	XML	3.98-1.3	2015-06-30
##	xml2	0.1.2	2015-09-01
##	xtable	1.7-4	2014-09-12
##	XVector	0.8.0	2015-04-17
##	yaml	2.1.13	2014-06-12
##	zlibbioc	1.14.0	2015-04-17
##	source		
##	CRAN (R 3.2.0)		
##	Bioconductor		
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##	Bioconductor		
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##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.2)		
##	Github (jdstorey/edge@a1947b5)		
##	CRAN (R 3.2.2)		
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##	CRAN (R 3.2.0)		
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##	CRAN (R 3.2.0)		
##	CRAN (R 3.2.1)		

```
## Bioconductor
## Bioconductor
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## local
## CRAN (R 3.2.0)
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## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
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## CRAN (R 3.2.1)
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## Bioconductor
## Github (alyssafranze/RSkittleBrewer@0a96a20)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.2)
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It is also useful to compile the time the document was processed. This document was processed on: 2015-09-06.